

Algorithms

Surface blasting design calculator

1. Start
2. Declare the variables Blasting Diameter (BD), Average Inhole Density (p), Spacing To Burden Ratio (S/B ratio) and Bench Height (BH).
3. "Ask the user to enter Blasting Diameter (BD)"
4. "Ask the user to enter Average Inhole density (p)"
5. "Ask the user to enter the Spacing to Burden ratio (S/B ratio)"
6. "Ask the user to enter the Bench Height (BH)"
7. Calculate Stemming length
8. Calculate Burden. Write a function for calculating a burden, that if the rock type is hard rock then the rock factor must be equal to 20 and soft rock factor equal to 35.
9. Calculate Actual powder factor
10. Display the results
11. Output
12. End

Flow diagram for Surface blasting design calculator

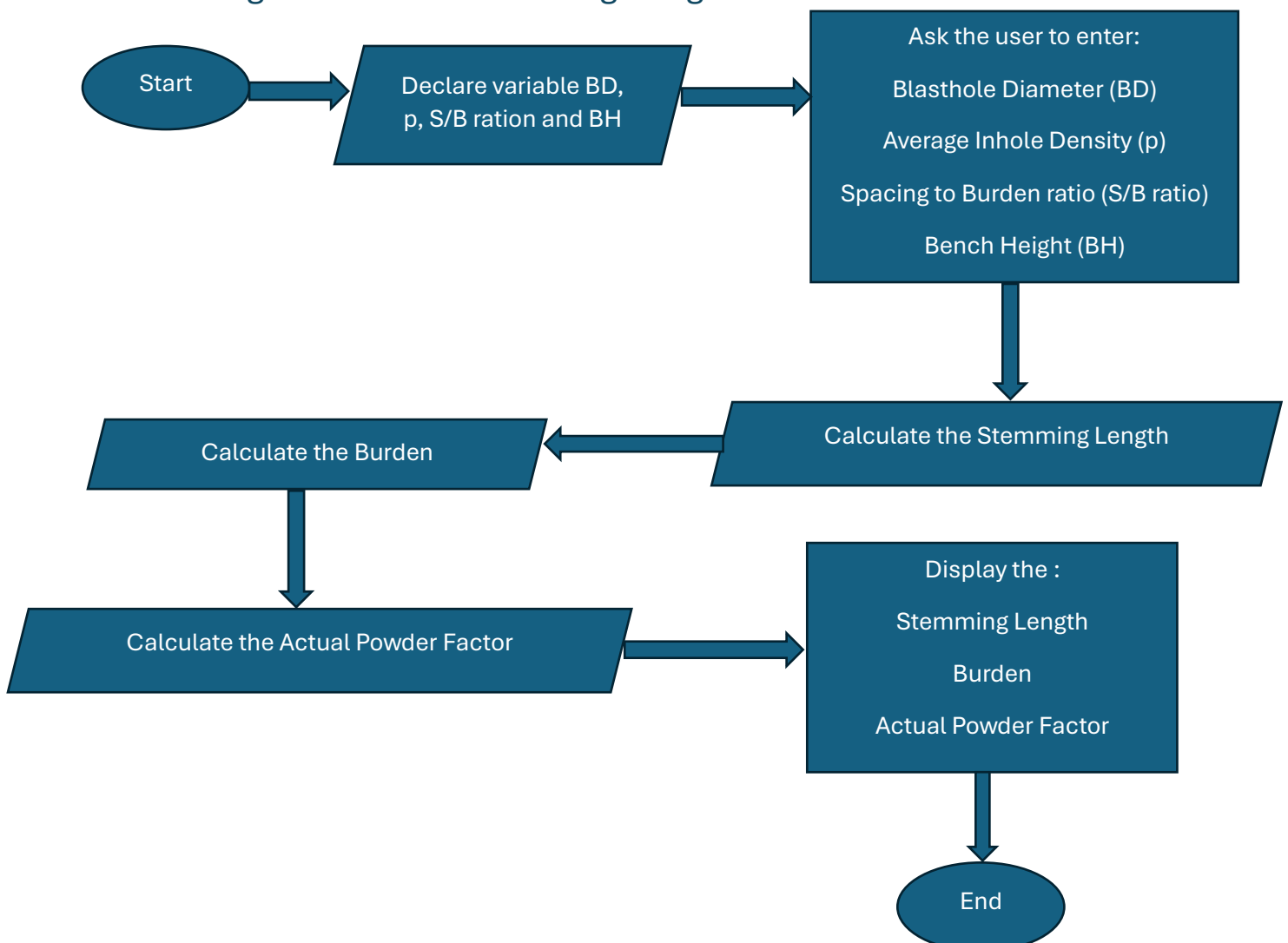


Figure 1: Flow Diagram for the Surface Blasting Design Calculator

Fragmentation Calculator

1. Start
2. Declare variables Average Inhole Density (p), Bench Height (BH), Relative effective Energy of Explosive (E) and Stemming Length
3. "Ask the user to enter Average Inhole Density (p)"
4. "Ask the user to enter Bench Height (BH)"
5. "Ask the user to enter Relative Effective Energy of Explosive (E)"
6. "Ask the user to enter Stemming Length"
7. Use Kuz-Ram Fragmentation Model Blasting Formula to Calculate the Mean Fragmentation Size.
8. Display the Mean Fragmentation Size.
9. Output
10. End

Flow diagram for Fragmentation Calculator

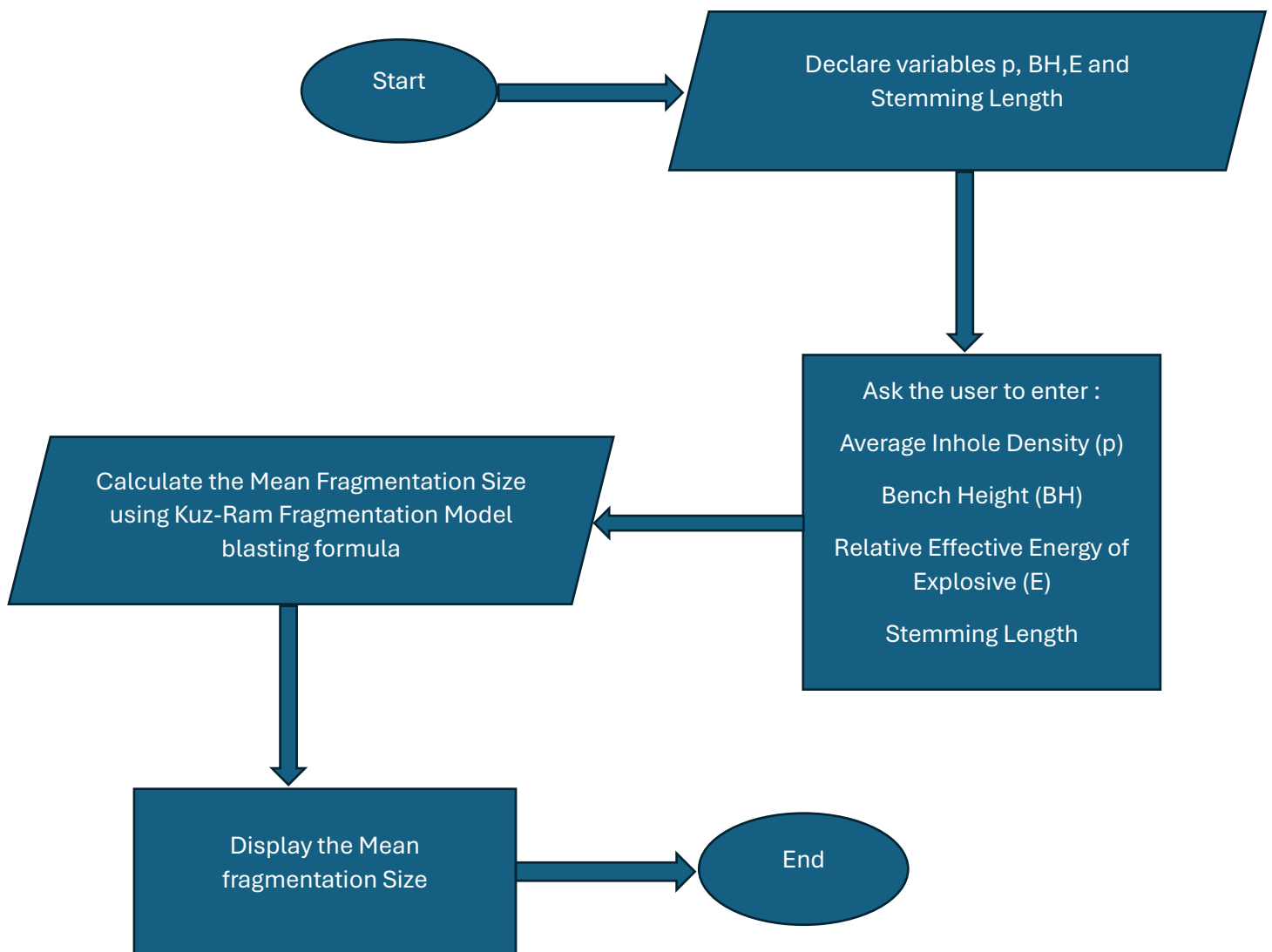


Figure 2: Flow diagram for the Fragmentation calculator